CLAIMS

What is claimed is:

1. A method for secure communication between a first device and a second device over a parallel communication channel, the method comprising the steps of:

- generating a session-specific data encoding map based on a shared secret known to the first and second devices;
- transmitting data from the first device to the second device, wherein the data is encoded into a sequence of physical channel states according to the session-specific map;
- receiving the physical channel states at the second device and decoding the data according to the session-specific map; and
- deleting the session-specific map on at least one of the devices upon termination of a communication session.

2. The method of claim 1, wherein the shared secret is a cryptographic seed, and the map is generated using a deterministic pseudo-random algorithm.

3. The method of claim 1, wherein the shared secret is exchanged via a secure out-of-band channel.

4. The method of claim 1, wherein the parallel communication channel has a cardinality greater than 2.

5. The method of claim 4, wherein the channel comprises a plurality of pixels, and the map creates a correspondence between multi-byte data blocks and unique pixel colors.

6. The method of claim 1, wherein an eavesdropper intercepting the sequence of physical channel states without the session-specific map is unable to decode the data.

7. The method of claim 1, wherein the parallel communication channel is a massively parallel optical interface as described in U.S. Patent Application No. 19/559,705.